

Segnet-Based Gland Segmentation from Colon Cancer Histology Images

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Abstract—The morphology of glands is the main basis of colon cancer diagnosis and accurate segmentation of glands from histology images is a prerequisite for correct clinical diagnosis. A gland segmentation method based on Segnet is proposed in this paper. First, the Warwick-QU dataset is used and augmented for training Segnet. Second, the network parameters are optimized according to the training results. Finally, the trained Segnet is used to perform segmentation test on both Parts A and B of Warwick-QU. The results show that the presented method achieves segmentation accuracy of 0.882 on Part A and 0.8636 on Part B, and shape similarity 106.6471 on Part A and 102.5729 on Part B, respectively. Compared with the existing methods with respect to the same dataset, our method reaches the highest accuracy on whole.

Keywords—Colon cancer, Deep learning, Gland segmentation, Histology image, Segnet.

I. INTRODUCTION

The incidence and mortality rate of colon cancer has increased in recent years. According to the 2017 U.S. Cancer report [1], the colon cancer is the second leading cause of death in men while the third in women. At present, the diagnosis of colon cancer mainly depends on the diagnosis of pathologists. Pathologists generally draw conclusions by reading pathological images, which is from puncture biopsies and section staining. This process not only takes a lot of time and cost, but also has obvious limitations. The research shows that the diagnosis of different pathologists has greater inconsistency [2][3]. The main reason for this inconsistency is that the artificial diagnosis method is subjective and easy to be affected by the environment. Nowadays, with the rapid development of computer technology and the popularization of pathological full-slice scanning equipment, it is possible to digitize pathological samples. The computer-aided diagnosis of digital pathological images has become one of the research hotspots in the field of medical images[4]-[7], and it can be an effective method for auxiliary diagnosis.

Gland is an important tissue structure in pathological images, it is essential to segment the gland for colon cancer computer-aided diagnosis. There are millions of glands in human colon and rectum which protect epithelial tissue and cells by absorbing water and nutrients and secreting mucus to provide a favorable basis for the regeneration of epithelial

cells [8][9]. However, the loss of integrity of epithelial cells in proliferation is one of the causes of colon cancer. Hence, accurate segmentation of glands is one crucial pre-requisite step to obtain reliable morphological statistics that indicate the aggressiveness of tumors. Recently, more and more scholars apply the method of deep learning to pathological image analysis, among which convolution neural network model is the most widely used [10]-[13]. Xu *et al.* [14] proposed a deep fully convolutional neural network for pixel-wise segmentation of epithelial and stromal in breast histology images. Ciresan *et al.* [15] use deep convolution neural network to detect mitosis in mammary tissue image and obtain the champion of ICPR2012 mitosis test contest. Chen *et al.* [16] proposed a deep-contour-aware networks for accurate gland segmentation in colon pathological images.

Segnet is a deep fully convolutional neural network architecture for pixel-wise segmentation [17]. It is an encoder-decoder network which is outperformed in semantic segmentation. Inspired by this, we proposed a gland segmentation of colon cancer histology image method based on Segnet, which brings a superior performance.

II. ACQUISITION AND AUGMENTATION OF DATASET

A. Warwick-QU Dataset

We evaluated our method on the public benchmark dataset of Gland Segmentation Challenge Contest in MICCAI 2015 (named as Warwick-QU dataset).

The images were acquired by a Zeiss MIRAX MIDI slide scanner from colon cancer tissues with a resolution of pixel. They consist of a wide range of histologic grades from benign to malignant subjects. The training dataset is composed of 85 images (benign/malignant=37/48) with ground truth annotations provided by expert pathologists. The testing data contains two sections: Part A (60 images) for offline evaluation and Part B (20 images) for on-site evaluation. Fig. 1 shows the examples of Warwick-QU dataset in benign and malignant cases.

The gland region is annotated by red line in Fig. 1, we can see that the gland structure is clear and is in normal size in benign images, which is dissimilated seriously in malignant images.

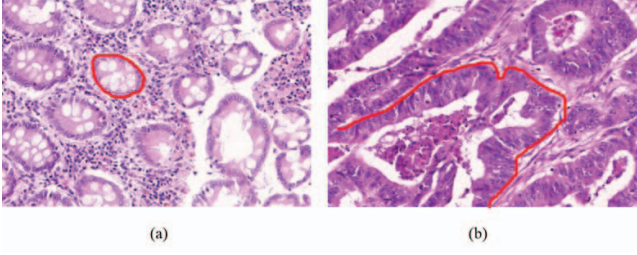


Fig. 1. Examples of benign (a) and malignant (b) cases.

B. Data Augmentation

To increase the robustness and reduce over-fitting, we utilized the strategy of data augmentation to enlarge the dataset. The common augmentation transformations include rotation, scale, random crop, noise and flip. We choose the random crop in this method, which means using random difference method to cut the image.

III. GLAND SEGMENTATION METHOD

A. Evaluation Index

The performance of this segmentation method is evaluated based on two criteria [18]: 1) accuracy of the segmentation of individual glands; 2) shape similarity between glands and their corresponding segmentation.

1) Segmentation Accuracy

The Dice index is a measure of agreement or similarity between two sets of samples [19]. Given G , a set of pixels belonging to a ground truth object, and S , a set of pixels belonging to a segmented object, the Dice index is defined as follows:

$$\text{Dice}(G, S) = \frac{2 * |G \cap S|}{|G| + |S|}, \quad (1)$$

where $|\cdot|$ denotes set cardinality. The index ranges over the interval $[0,1]$, where the higher the value, the more concordant the segmentation result and the ground truth. A Dice index of 1 implies a perfect agreement. The calculation made in this way measures the segmentation accuracy only at the pixel level, not at the gland level, which is not a perfect way to evaluate the segmentation performance.

To take the notion of an individual gland into account, we employ the object-level Dice index and it is defined as [20]:

$$\text{Dice}_{obj}(G, S) = \frac{1}{2} * \left[\sum_{i=1}^{n_G} \gamma_i * \text{Dice}(G_i, S_*(G_i)) + \sum_{j=1}^{n_S} \sigma_j * \text{Dice}(G_*(S_j), S_j) \right] \quad (2)$$

$$\gamma_i = |G_i| / \left| \sum_{p=1}^{n_G} |G_p| \right|, \sigma_j = |S_j| / \left| \sum_{q=1}^{n_S} |S_q| \right|, \quad (3)$$

where n_G denotes the number of non-empty ground truth gland, as annotated by the expert pathologist. Similarly n_S denotes the number of glands segmented by the segmentation method, that is the number of non-empty

segmented objects. G_i denotes the i^{th} ground truth object, and S_j denotes the j^{th} segmented object. $S_*(G_i)$ denotes a ground truth object that maximally overlaps G_i , $G_*(S_j)$ denotes a segmented object that maximally overlaps S_j .

2) Shape Similarity

The shape similarity is measured by using the object-level Hausdorff distance between the shape of segmented object in S and that of the ground truth object in G . The usual definition of a Hausdorff distance between ground truth object G and segmented object S is

$$H(G, S) = \max \left\{ \sup_{x \in G} \inf_{y \in S} d(x, y), \sup_{y \in S} \inf_{x \in G} d(x, y) \right\}, \quad (4)$$

where $d(x, y)$ denotes the Euclidean distance between pixels $x \in G$ and $y \in S$.

Hausdorff distance is the most extreme value from all distances between the pairs of nearest pixels on the boundaries of S and G . Thus, the smaller the value of the Hausdorff distance, the higher the similarity between the boundaries of S and G , and $S = G$ if their Hausdorff distance is zero.

Likewise, an object-level Hausdorff is employed:

$$H_{obj}(G, S) = \frac{1}{2} * \left[\sum_{i=1}^{n_G} \gamma_i * H(G_i, S_*(G_i)) + \sum_{j=1}^{n_S} \sigma_j * H(G_*(S_j), S_j) \right], \quad (5)$$

where the meaning of the mathematical notation is similar to that given in Dice_{obj}

B. Configuration and Training of Segnet

1) Segnet

The proposed method of gland segmentation in colon Cancer Histology Images is to segment the structure of glands in pathological images using deep convolution encoder-decoder network.

Segnet has a encoding network and a corresponding decoding network, followed by a final pixelwise classification layer. The encoding network consists of 13 convolutional layers which designed for object classification. Each encoder layer has a corresponding decoder layer and hence the decoder network has 13 layers. The final decoder output is fed to a multi-class softmax classifier (SMC) to produce class probabilities for each pixel independently.

Each encoder in the encoder network performs convolution with a filter bank to produce a set of feature maps. These are then batch normalized [21][22]. Then an element-wise rectifiedlinear non-linearity (ReLU) $\max(0, x)$ is applied. Following that, max-pooling is performed and the resulting output is sub-sampled.

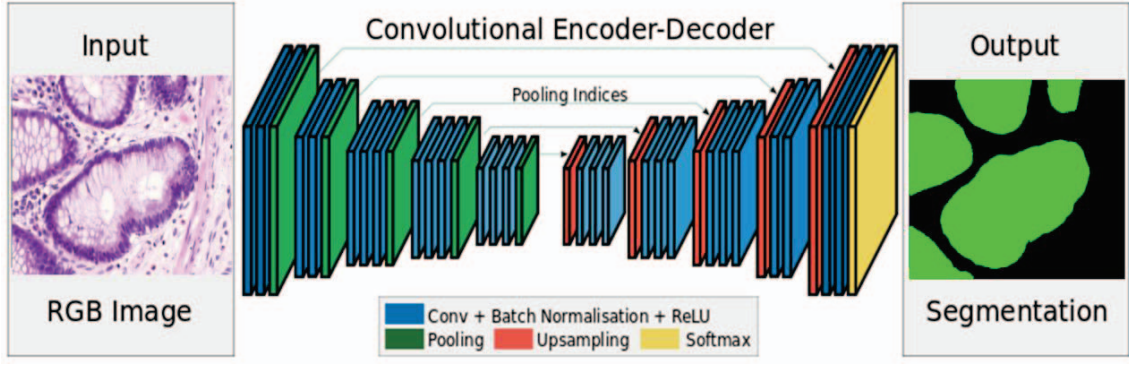


Fig. 2. Overview of Segnet

The appropriate decoder in the decoder network upsamples its input feature maps using the memorized max-pooling indices from the corresponding encoder feature maps. These feature maps are then convolved with a trainable decoder filter bank to produce dense feature maps. A batch normalization step is then applied to each of these maps. The high dimensional feature is fed to a trainable softmax classifier, which classifies each pixel independently. The output of softmax is a K channel image of probabilities where K is the number of classes. The predicted segmentation corresponds to the class with maximum probability at each pixel.

The convolutional layer can be described as follow:

Let $W^l = (w_1^l, w_2^l, \dots, w_{d_h^l}^l)$ be a filter bank. Each w_k^l , $k \in \{1, 2, \dots, d_h^l\}$ is an $m^l \times m^l$ linear filter embedded in the l -th layer. d_h^l is the number of different filters or kernels in the filter bank W^l . An input $\omega^{l-1} \times \omega^{l-1}$ patch I_p^{l-1} is convolved with a $m^l \times m^l$ local respective region in the image I_p^{l-1} encompassed by the filter w_k^l . The filter w_k^l moves over the input patch I_p^l and allows for a local convolution operation. For each patch extracted from the image, the convolution operations with d_h^l filters result in d_h^l feature maps. The result of the linear convolutional operation on I_p^{l-1} by w_k^l can be re-written as $g_k^l = I_p^{l-1} * w_k^l$. Here the pixel value $g_k(p)$ of each location of resultant image g_k^l is the sum of products of the filter coefficients and the image pixels of the local respective region encompassed by filter. During the convolutional operations, each filter w_k convolves across $(\omega^{l-1} - m^l + 1) \times (\omega^{l-1} - m^l + 1)$ pixels in I_p^{l-1} . The output of active function is

$$x^l = f(g_k^l) = \frac{1}{1 + \exp(-g_k^l)} \quad (6)$$

The max-pooling layers aim to achieve spatial invariance by reducing the resolution of the feature maps obtained in convolutional layers. It applies local pooling of feature maps using a max operation the neighbourhood of the results of the convolutional layers. Subsequent to the max-pooling operation, the size of the resultant images becomes $\frac{(\omega^{l-1} - m^l + 1)}{s} \times \frac{(\omega^{l-1} - m^l + 1)}{s}$, where s refers to the scale at which the operation is applied.

Softmax classifier is a supervised model generalizes logistic regression as

$$p_{W^{(l)}}(y^l = i | x^l) = \frac{1}{1 + \exp(-W^{(l)T} x^l)}, \quad (7)$$

where $p_{W^{(l)}}(\bullet) (i \in \{0, 1\})$ is a sigmoid function with parameters $W^{(l)}$. The input $x^{(l)}$ of SMC is a high-level feature learned by the Segnet. The parameter $W^{(l)}$ corresponding to the SMC learned via a training set. With the learned parameter $W^{(l)}$, each patch that is fed to the function produces a value between 0 and 1 that can be interpreted as the probability of the input patch corresponding to the gland region and background. The predicted class $\hat{i}$ and prediction score $\hat{s}(\hat{i})$ can then be represented as

$$\hat{i} = \left\{ i \mid \max_i p_{W^{(l)}}(\hat{y} = i) \right\}, \quad i \in \{0, 1\} \quad (8)$$

and

$$\hat{s}(\hat{i}) = p_{W^{(l)}}(y^l = \hat{i}), \quad \hat{i} \in \{0, 1\}, \quad (9)$$

respectively.

2) Network Training

First of all, the training data set and its ground truth are prepared for the training process, then the corresponding label files are generated respectively. After configuring the Segnet network structure, the training set and network structure are called by the script file to start the training process. During the process of training, the training performance is optimized by adjusting the internal parameters of the network. The training can be finished and

the network model can be obtained when the training accuracy and loss tend to be stable.

IV. SEGMENTATION TEST AND RESULT ANALYSIS

A. Experiment

All experiments were carried out on a PC with 2.4GHz Intel(R) Xeon(R) E5-2620 v3 CPU (32GB of RAM) and a NVIDIA GeForce GTX 1080 GPU. The software implementation was performed by Python 2.7 on Caffe framework [23].

During the training process of Segnet, there are some parameters which are set up before training. All the kernel size for convolutional layer is 3×3 and 2×2 for max-pooling layer. The learning rate for training is 0.001 and the policy of learning is “step”, which means the learning rate is changing according to the training performance.

According to the introduction of data augmentation, in order to ensure that every image contains clear gland structure and the network model has a superior training performance, we crop 5 regions with size of 480×360 from the original image as input data set after many comparative experiments. Hence there are 425 training image and 400 testing image after cropping. Likewise, the ground truth is cropped the same size. The method shows in Fig. 3.

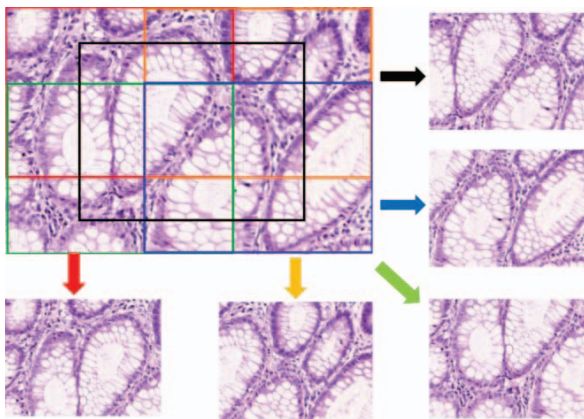


Fig. 3. Cropped images

We obtained the trained network model by training Segnet with 425 training images and ground truths, then tested this network model with the 400 testing images. The qualitative segmentation results are shown in Fig. 4.

In order to illustrate the efficacy of our method quantitatively, the evaluation criteria includes object-level Dice and Hausdorff distance, which consider the performance of segmentation accuracy and shape similarity, respectively. The segmentation accuracy and shape similarity results of the proposed method and method in this challenge are shown in Table I.

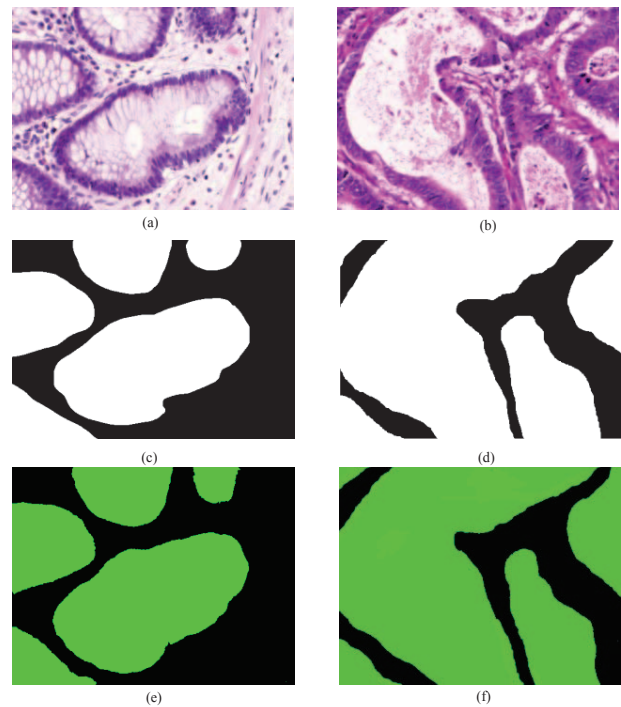


Fig. 4. Segmentation results of benign and malignant cases. (a) and (b) are pathological image, resp. (c) and (d) are ground truths and the gland (white) and background (black) are annotated from expert pathologists, resp. (e) and (f) are segmentation results of Segnet, i.e., the green region means segmented gland structures, the black region means background, resp.

TABLE I. TSET RESULTS

Method	Part A		Part B	
	Dice _{obj}	H _{obj}	Dice _{obj}	H _{obj}
CUM1	0.8666	74.5955	0.8001	153.646
CUM2	0.8974	45.4182	0.781	160.347
ExB1	0.8823	57.4126	0.786	145.575
ExB2	0.8844	54.7853	0.7542	187.442
ExB3	0.886	57.35	0.7647	159.873
Fre1	0.8745	57.1938	0.7832	146.607
Fre2	0.8756	57.0932	0.7647	148.463
Dund	0.8698	58.3386	0.7152	209.048
CVML	0.6444	155.4326	0.6543	176.244
LIB	0.8012	101.1668	0.6166	190.447
LIST	0.5789	239.0144	0.6251	222.723
V4G1	0.7374	107.4907	0.4357	413.934
v4G2	0.7374	107.4907	0.6098	210.105
Bio	0.5049	264.7235	0.6532	213.719
SUTH	0.5813	185.1403	0.4954	263.431
ISI	0.4924	157.0522	0.4436	216.17
FIMM	0.6092	244.3978	0.5518	330.232
Wan1	0.5607	251.1647	0.6139	278.919
Wan2	0.5423	252.4583	0.6052	278.261
Wan3	0.4351	324.1298	0.5684	300.71
Segnet	0.8820	106.6471	0.8636	102.5729

B. Result Analysis

From the segmentation results in Fig. 4 we can clearly see that the glands in benign images are in normal size and uniform distribution. While the glands in malignant images dissimilated seriously, the distance between two glands is too dense that it is easy to be clustered together. However, the Segnet is capable of separating these touching gland objects clearly.

According to the introduction of two segmentation indexes, the segmentation accuracy and shape similarity results of different method are shown in Table I. The segmentation accuracy on Part A has achieved 0.8820, which is a little lower than the best performance of 0.8974. Moreover, the shape similarity on Part A is 106.6471, which is on the forefront of the list. As for Part B, our method has achieved the best performance with the segmentation accuracy and shape similarity of 0.8636 and 102.5729, respectively, which outperformed all the other advanced methods. In summary, the proposed method has achieved the highest accuracy on whole.

V. CONCLUSIONS

In this paper, we have presented a Segnet-based segmentaion method for accurate segmenting the gland structures in colon cancer histology images. First, the training dataset for Segnet model is obtained by data augmentation. Second, the network parameters are adjusted for optimizing network performance according to training results. Third, the segmentation performance is tested on the dataset by using the trained network model. Finally, we evaluate the segmentation results in terms of segmentation accuracy and shape similarity. Our results show that the proposed method can achieve the highest accuracy on whole. Our work also suggests that the deep learning methods including Segnet are promising for the auxiliary diagnosis of pathologists.

ACKNOWLEDGEMENT

This work was supported in part by the National Natural Science Foundation of China under Grant 61773115 and the Jiangsu Province Natural Science Foundation under Grant BK20161427. Also, we thank Professor Jun Xu and his Ph.D. student Chao Zhou with Jiangsu Key Laboratory of Big Data Analysis Technique, Nanjing University of Information Science and Technology, for their constructive suggestions and unselfish assistance.

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